

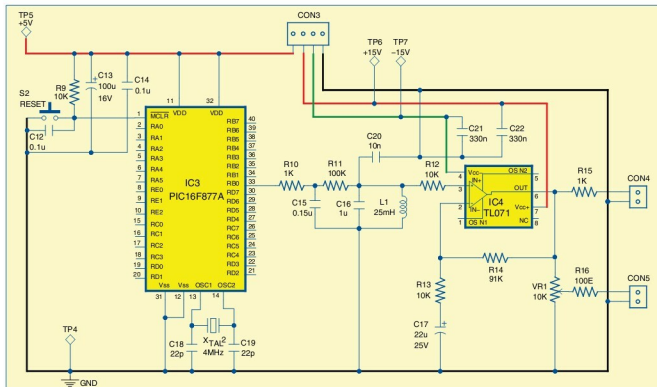


Fig. 2: Circuit diagram for producing a sinusoidal signal with a combination of RC and LC filters

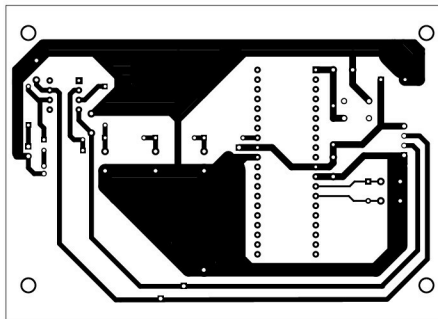


Fig. 3: Actual-size PCB pattern of the circuit for a sinusoidal signal with RC filters

ducing a sinusoidal signal using a combination of RC and LC filters for better quality of filtration.

The RC filter is built with R10 and C15. Its cut-off frequency is equal to:

$$F(-3dB) = 1/(2 \times \pi \times R10 \times C15) = 1/(6.28 \times R10 \times C15) [Hz]$$

The LC filter is built with L1 and C16. Its resonant frequency is equal to:

$$Fr = 1/(2 \times \pi \times (\text{sqrt}(L1 \times C16))) [Hz]$$

 Ideally, $F(-3dB) = Fr$, but in practice, an accuracy of $\pm 5\%$ or even $\pm 10\%$ is acceptable.

PARTS LIST

Semiconductors:

- IC1, IC3 - PIC16F877A 8-bit MCU
- IC2, IC4 - TL071 low-noise JFET-input operational amplifier

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R3, R5, R6, R9, R12, R13 - 10-kilo-ohm
- R2, R10, R15 - 1-kilo-ohm
- R4, R11 - 100-kilo-ohm
- R7, R14 - 91-kilo-ohm
- R8, R16 - 100-ohm
- VR1 - 10-kilo-ohm potmeter

Capacitors:

- C1, C3, C12, C14 - 0.1 μ F ceramic disk
- C2, C13 - 100 μ F, 16V electrolytic
- C4, C15 - 0.15 μ F ceramic disk
- C5 - 15nF ceramic disk
- C6 - 1.5nF ceramic disk
- C7, C17 - 22 μ F, 25V electrolytic
- C8, C9, C18, C19 - 22pF ceramic disk
- C16 - 1 μ F, 25V MLCC
- C20 - 10nF ceramic disk
- C10, C11, C21, C22 - 330nF ceramic disk

Miscellaneous:

- CON1, CON3 - 4-pin connector
- CON2, CON4, CON5 - 2-pin connector
- L1 - 25mH inductor
- S1, S2 - Tactile switch
- XTAL1, XTAL2 - 4MHz crystal

Values of the components are given for producing a sinusoidal signal of 1kHz. Capacitor C20 can be added